Teaching Statement

Jane Hsieh

As a teacher and mentor, I strive to create approachable settings for a wide range of students to apply, internalize and deliberate on knowledge with sufficient context for application and space for creativity. To achieve this in a rapidly evolving world, I continuously create and adapt both digital and classroom resources that motivate students to engage in ① active learning and ② collaborative feedback ③ so they build intuition and independent, critical thinking to navigate accelerating change. My background has prepared me to teach undergraduate courses in human-AI interaction, software engineering, data structures and algorithms as well as graduate-level courses covering special topics on the future of work and technology, user-centered design and computational social science.

Teaching

In graduate school, I was the sole TA for two iterations of Introduction to Human AI Interaction (HAII) – which ranges ~50-00 students, and Introduction to Software Construction – an invite-only course targeted to improve retention rate for a core CS pre-requisite. In these experiences I created syllabi, prepared and delivered lectures, managed homeworks and designed policies around AI use. During my undergraduate degree, I served as a teaching assistant for three courses, where I grew in a variety of TA (including lecturing) roles in a liberal arts setting. I continuously seek opportunities to hone my teaching and mentoring, such as by taking courses in CS Pedagogy (CMU), Language Pedagogy (Oberlin) and Graduate Student Mentor Training.

Guiding Active and Applied Learning

> *Knowledge is not a copy of reality [...] To know an object is to act on it [...] to transform the object, and to understand the process of this transformation [3]*

As researchers and knowledge producers, we are experts in creating and delivering accurate and up-to-date course material, yet student engagement with course content rely primarily on relevance and their perceived subjective value. To incentivize hands-on engagement with content-heavy lectures in HAII, I integrated structured pre-class readings and discussions to guide students in developing their own perspectives. At the end of each class session, I led collaborative classroom exercises to encourage active application of lecture material through divergent and explorative thinking. As an example, students in the first iteration of HAII thoroughly immersed themselves in the Black Mirror Writers' Room — an in-class activity based on the exercise by Fiesler to speculate on the harms of future technologies. But despite their investment reflecting on ethics and societal impacts of AI, I worried how this critical lens might foster a fear of advancements. Thus, when I redesigned course materials, I included a complementary "*Light Mirror*" exercise where students develop second-episode plots – directing them to look beyond the problem space and practice devising and critiquing their own (alternative) solutions. To encourage in-depth engagement in future teaching, I will incorporate exercises (*e.g.* socratic debates) that require more critical and reflective thinking.

Collaborative, Creative and Constructive Peer Feedback

> *The* "deep secret" *of collaborative learning seems to lie in the cognitive processes through which humans progressively build a shared understanding* [4]

Beyond course content, *how* students engage with material also shapes their learning. I design collaborative spaces and activities where students can understand, critique and build off of each other's insights, reflecting realistic contexts where interdisciplinary team members work towards shifting goals [5]. In office hours for introductory algorithms, I observed that students who actively helped peers solve a problem not only found the experience rewarding, but also strengthened their own conceptual understanding by seeing the solution from another perspective . More digitally, I prepare each course with online spaces (*e.g.* Piazza, Slack) to support asynchronous collaboration. Finally, I use stepwise, critical questioning in small group contexts to help students build on prior knowledge, and plan to design activities in future courses to foster similar styles of grounded and collaborative learning.

Fostering Growth in Evolving Landscapes

> *To work toward a just world – a world where all have equal access to opportunity – means, as a start, opening up heart and mind to the perspectives of others. [1]*

As educators, we must continuously adapt to ensure our content and methods align with student learning goals, leveraging both available and emerging resources. During my second term as a TA for HAII (Spring '23), students voiced intense and valid concerns about job displacement from recent LLM advances. To address such genuine underlying fears, I incorporated material contextualizing the past, present, and potential futures of computing, highlighting how AI is likely to create more creative and knowledge-based roles, much as manufacturing once absorbed agricultural workers in the First Industrial Revolution. Transitioning to the present, I added lectures on the evolution of AI writing support – from early spell-check to modern generative tools – and approaches for detecting and mitigating AI failures. Looking to the future, I introduced speculative design methods like speed dating and storyboarding to give students tools to explore alternatives. I also participated in the workshop *Adapting to Generative AI in the Classroom* at CMU's Eberly Center to consider new ways of assessment and material development – leading to the co-creation of AI policies with students. Moving forward, I am committed to situating students and ourselves within the material we deliver to improve content relevance and cultivate more well-rounded student perspectives and intuition.

Mentoring

As a mentor, I guide students to develop interpersonal and research skills, explore interests, and provide resources and direction for projects over three stages. My 13 mentees (11 undergraduate, 2 masters) from varied backgrounds (9 different institutions) imparted me with valuable, cross-disciplinary collaborations and perspectives. Two of my past mentees are now pursuing PhDs of their own at UPenn and NYU.

Defined Beginnings: To start, I outline for mentees higher-level objectives and concrete weekly milestones to clearly communicate expectations in writing. This practice, inspired by the Independent Development Plan I developed as a part of the Undergraduate Research Engagement Working Group, began with my first pair of summer students in 2022. Beyond onboarding and short-term goals, I also provide readings and resources within these documents to orient mentees and contextualize their roles in prior research.

***Feedback-based Execution*:** While short-term goals guide individual tasks, longer-term aims (*e.g.* project objectives, career ambitions) require more iterative exploration and refinement. To support *research skill* development, I provide incremental feedback on how each mentee synthesizes, communicates, and builds upon research findings through guided milestones such as literature surveys, stand-ups, and retrospective syncs. *Interpersonally*, I encourage mentees to raise concerns about project directions or technical struggles as early as possible, to build and normalize practices of actively seeking guidance. In a recent game design project, one mentee repeatedly emphasized immersion as a key mechanism – challenging and expanding my understanding on the persuasive potentials of gamified interventions, in addition to broadening the overall project scope.

Guided Reflections & Future Directions**:** Over the course of the project, I schedule mid-summer check-ins to continually assess mentees' expectations, as well as end-of-week check-ups about weekend plans (reverse ice-breakers) to help transition between work to life. After their designated time with us, 7 of 9 summer students volunteered to continue researching, where they wrapped up final pieces and reflexively oriented toward their long-term interests. Beyond supporting my own students, I also co-organized for CMU's graduate application support program, which scalably matches aspiring applicants with current computing CMU computing PhDs to give feedback on statements – reaching more than 100 external applicants in 2024 alone. As a faculty member, I intend to implement similar programs to bridge gaps between student communities, particularly those of more historically underrepresented backgrounds.

Reference

[1] Takacs, D. *Positionality, Epistemology, and Social Justice in the Classroom.* Pedagogies for Social Change, 2002

[2] Ambrose, S. A., Bridges, M. W., DiPietro, M., Lovett, M. C., & Norman, M. K. *How learning works: Seven research-based principles for smart teaching*. John Wiley & Sons, 2010.

[3] Piaget, J. (1964). *Part I: Cognitive development in children: Piaget. Development and learning.* Journal of Research in Science Teaching, 2(3), 176–186.

[4] Weiss, G., & Dillenbourg, P. (1999). What is 'multi' in multi-agent learning? In P. Dillenbourg (Ed.), *Collaborative Learning: Cognitive and Computational Approaches* (pp. 64–80).

[5] Nahar, N., Zhou, S., Lewis, G., & Kästner, C. (2022, May). Collaboration challenges in building ml-enabled systems: Communication, documentation, engineering, and process.